

CS 228 : Logic in Computer Science

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Recap

- ▶ Herbrand universe on a signature τ
- ▶ Herbrand structure

FO without equality

Let us focus on FO without “=”. Recall that “=” is always interpreted as equality.

Herbrand Theorem

Let $\Gamma = \{\varphi_1, \varphi_2, \dots\}$ be a set of equality-free sentences in Skolem Normal Form. Then Γ is satisfiable iff Γ has a Herbrand model.

If Γ has a Herbrand model, clearly Γ is satisfiable. The converse needs a proof.

The converse

Assume Γ is satisfiable. Let τ_H be the Herbrand signature for Γ .

- ▶ Let $\mathcal{A}$ be a τ_H structure such that $\mathcal{A} \models \Gamma$. ($U^{\mathcal{A}}$ need not be the Herbrand universe)
- ▶ We need to create a Herbrand model for Γ

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- ▶ Let R be an n -ary relation in τ_H (hence in τ).
- ▶ For each n -tuple $(t_1, \dots, t_n)$ with t_i coming from the Herbrand universe $H(\Gamma)$, we must say whether $(t_1, \dots, t_n) \in R^M$ or not
- ▶ Each $t_i \in H(\Gamma)$ is a ground term in τ_H (so variable free).
- ▶ Since $\mathcal{A}$ is a structure over τ_H , if $t \in H(\Gamma)$ is a ground term from τ_H , $\mathcal{A}$ interprets t as an element of $U^{\mathcal{A}}$.
- ▶ For each n -tuple $(t_1, \dots, t_n)$, $t_i \in H(\Gamma)$, we know whether $(t_1, \dots, t_n) \in R^{\mathcal{A}}$ or not
- ▶ Define $(t_1, \dots, t_n) \in R^M$ iff $(t_1, \dots, t_n) \in R^{\mathcal{A}}$. (why does this suffice?)
- ▶ Prove that if $\mathcal{A} \models \varphi$ for any $\varphi \in \Gamma$, then $M \models \varphi$.
- ▶ The proof is by induction on the number of quantifiers in φ . Recall that each φ is in Skolem Normal Form.

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- ▶ Assume φ is an atomic formula. (what are they?)
- ▶ Then φ is $R(t_1, \dots, t_n)$ where R is an n -ary relation from τ_H , and $t_1, \dots, t_n$ are all terms from $H(\Gamma)$.
- ▶ By the construction of M , $R^M = R^{\mathcal{A}}$.
- ▶ Hence $M \models \varphi$ iff $\mathcal{A} \models \varphi$.
- ▶ Same reasoning holds for $\varphi_1 \wedge \varphi_2$, $\varphi_1 \vee \varphi_2$ and $\neg \varphi$.
- ▶ Hence, $\mathcal{A} \models \varphi$ iff $M \models \varphi$.

Post Inductive Hypothesis

Assume that for any $\psi \in \Gamma$ with $\leq k - 1$ quantifiers, if $\mathcal{A} \models \psi$, then $M \models \psi$. Let $\varphi \in \Gamma$ have k quantifiers, $\varphi = \forall x_1 \forall x_2 \dots \forall x_k \zeta(x_1, \dots, x_k)$ where ζ is quantifier free.

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- ▶ Let $\kappa(x_1) = \forall x_2 \dots \forall x_k \zeta(x_1, \dots, x_k)$, and $\varphi = \forall x_1 \kappa(x_1)$.
- ▶ $\mathcal{A} \models \varphi$ implies $\mathcal{A} \models \forall x_1 \kappa(x_1)$. That is, $\mathcal{A} \models \kappa(a)$ for any $a \in U^{\mathcal{A}}$.

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- ▶ Since $\mathcal{A}$ is a structure over τ_H , if $t \in H(\Gamma)$ is a ground term from τ_H , $\mathcal{A}$ interprets t as an element of $U^{\mathcal{A}}$.
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- ▶ Thus, $\mathcal{A} \models \kappa(t)$ for any $t \in H(\Gamma)$.
- ▶ By induction hypothesis, $M \models \kappa(t)$ for any $t \in H(\Gamma)$.
- ▶ Since $H(\Gamma)$ is the universe of M , $M \models \forall x_1 \kappa(x_1)$. That is, $M \models \varphi$.

The Herbrand Theorem follows.

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- ▶ However, φ is satisfiable. Define a structure $\mathcal{A} = (\{0, 1\}, f^{\mathcal{A}}(0) = 1, f^{\mathcal{A}}(1) = 0)$, $\mathcal{A} \models \varphi$
- ▶ For formulae which have equality, Herbrand's Theorem does not apply directly
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Let φ be in Skolem normal form with equality. Then φ is satisfiable iff there is an equisatisfiable formula ψ in Skolem normal form without equality which has a Herbrand model.

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- ▶ Define φ_{ER} to be the sentence

$$\forall x \forall y \forall z (E(x, x) \wedge ((E(x, y) \leftrightarrow E(y, x)) \wedge (E(x, y) \wedge E(y, z) \rightarrow E(x, z))))$$

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- ▶ For each n -ary relation R in τ , define φ_R as

$$\forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n ((\bigwedge_{i=1}^n E(x_i, y_i) \wedge R(x_1, \dots, x_n)) \rightarrow R(y_1, \dots, y_n))$$

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- ▶ Let $\varphi_1 = \bigwedge_{R \in \tau} \varphi_R$

Dealing with Equality

- ▶ For each n -ary function f in τ , define φ_f as

$$\forall x_1 \dots \forall x_n \forall y_1 \dots \forall y_n \left(\left(\bigwedge_{i=1}^n E(x_i, y_i) \rightarrow E(f(x_1, \dots, x_n), f(y_1, \dots, y_n)) \right) \right)$$

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- ▶ Let $\varphi_2 = \bigwedge_{f \in \tau} \varphi_f$
- ▶ Let $\psi_E = \varphi_{\neq} \wedge \varphi_{ER} \wedge \varphi_1 \wedge \varphi_2$
- ▶ Convert ψ_E to Prenex normal form to obtain φ_E in Skolem normal form

For any formula φ in Skolem normal form, φ is satisfiable iff φ_E is satisfiable

Key Ideas for the Proof

- ▶ Let τ be the signature of φ and let $\tau_E = \tau \cup \{E\}$, where E is a binary relation not in τ .
- ▶ If $\mathcal{A} \models \varphi$ for some structure, we can obtain a model $\mathcal{A}^E$ for φ_E by interpreting E as equality, extending $\mathcal{A}$.

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- ▶ Conversely suppose φ_E is satisfiable, and let $\mathcal{B}^E \models \varphi_E$ for some structure $\mathcal{B}^E$.
- ▶ Note that E is an equivalence relation on the universe of $\mathcal{B}^E$.
- ▶ Let $C_{\mathcal{B}^E}$ be all equivalence classes in $\mathcal{B}^E$ induced by E .
- ▶ Define a structure $\mathcal{B}$ to satisfy φ

Key Ideas for the Proof

- ▶ Define structure $\mathcal{B}$ having its universe $\mathcal{U}^{\mathcal{B}}$ as $C_{\mathcal{B}^E}$
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 - ▶ Let f be an n -ary function in τ . For any $[b] \in \mathcal{U}^{\mathcal{B}}$ define $f^{\mathcal{B}}([a_1], \dots, [a_n]) = [b]$ if $f^{\mathcal{B}^E}(a_1, \dots, a_n) = b$

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 - ▶ Let f be an n -ary function in τ . For any $[b] \in \mathcal{U}^{\mathcal{B}}$ define
$$f^{\mathcal{B}}([a_1], \dots, [a_n]) = [b] \text{ if } f^{\mathcal{B}^E}(a_1, \dots, a_n) = b$$
- ▶ $\mathcal{B}$ is well defined since $\mathcal{B}^E$ satisfies φ_1 and φ_2
- ▶ Prove by structural induction that $\mathcal{B} \models \varphi$.

An Example

$$\varphi = \forall x[(f(x) \neq x) \wedge (f(f(x)) = x)].$$

- ▶ φ is satisfiable : $\mathcal{A} = (\{0, 1\}, f^{\mathcal{A}}(0) = 1, f^{\mathcal{A}}(1) = 0)$ and $\mathcal{A} \models \varphi$.
- ▶ $\varphi_{\neq} = \forall x[\neg E(f(x), x) \wedge E(f(f(x)), x)]$
- ▶ $\varphi_2 = \forall x \forall y[E(x, y) \rightarrow E(f(x), f(y))]$
- ▶ Conjoin $\varphi_{\neq}, \varphi_2$ and φ_{ER} and convert to Prenex normal form
- ▶ $\varphi_E = \forall x \forall y \forall z[(\neg E(f(x), x) \wedge E(f(f(x)), x)) \wedge$
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 $(E(x, y) \wedge E(y, z) \rightarrow E(x, z))]$
- ▶ By Herbrand's Theorem, φ_E has a Herbrand model M

An Example

- ▶ What are the equivalence classes induced by E on $H(\varphi)$ (the Herbrand universe of φ)?
 - ▶ $\text{Odd} = \{t \in H(\varphi) \mid t \text{ has odd number of } f\text{'s}\}$
 - ▶ $\text{Even} = \{t \in H(\varphi) \mid t \text{ has even number of } f\text{'s}\}$
- ▶ $M = (\{c, f(c), f(f(c)), \dots\}, E^M = \{(t, t') \mid \text{the number of } f\text{'s in both } t, t' \text{ have the same parity}\})$
- ▶ $M \models \varphi_E$

Herbrand's Method

Given a FO sentence φ , is it satisfiable? Wlg, assume that φ is equality-free and is in Skolem normal form.

- ▶ Let $\varphi = \forall x_1 \dots \forall x_n \psi(x_1, \dots, x_n)$
- ▶ $\psi(x_1, \dots, x_n)$ is quantifier-free, and can be written as a conjunction of disjunction of atomic formulas and negation of atomic formulas
- ▶ Let $H(\varphi)$ be the Herbrand universe of φ
- ▶ Let $E(\varphi) = \{\psi(t_1, \dots, t_n) \mid t_1, \dots, t_n \in H(\varphi)\}$ be the set obtained by substituting terms from $H(\varphi)$ for the variables $x_1, \dots, x_n$ in φ
- ▶ φ is satisfiable iff $E(\varphi)$ is satisfiable

Herbrand's Method

- ▶ Assume φ is satisfiable. Then $\mathcal{A} \models \forall x_1, \dots, x_n \psi(x_1, \dots, x_n)$
- ▶ In particular, $\mathcal{A} \models \psi(t_1, \dots, t_n)$ where $t_1, \dots, t_n \in H(\varphi)$
- ▶ Then $\mathcal{A} \models \varphi_i$ for all $\varphi_i \in E(\varphi)$
- ▶ Hence, $E(\varphi)$ is satisfiable.

Herbrand's Method

- ▶ Assume $E(\varphi)$ is satisfiable. $E(\varphi)$ is a set of equality-free sentences.
- ▶ By Herbrand's Theorem, there is a Herbrand model M for $E(\varphi)$.
- ▶ The Herbrand signature for formulae in $E(\varphi)$ is the same as the Herbrand signature of formula φ .
- ▶ The universe of M is $H(\varphi)$. For each $t_1, \dots, t_n \in H(\varphi)$, $M \models \psi(t_1, \dots, t_n)$ since $\psi(t_1, \dots, t_n) \in E(\varphi)$.
- ▶ By semantics of $\forall$, $M \models \forall x_1 \dots x_n \psi(x_1, \dots, x_n)$
- ▶ Then $M \models \varphi$ and φ is satisfiable.
- ▶ φ is unsatisfiable iff $E(\varphi)$ is unsatisfiable.

Checking Unsatisfiability of φ

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- ▶ $E(\varphi) = \{\varphi_1, \varphi_2, \dots\}$ is a set of quantifier free sentences, so it can be seen as a set of propositional logic formulae
- ▶ Each formula $\varphi_i \in E(\varphi)$ is in CNF
- ▶ We know that $E(\varphi)$ is unsatisfiable iff $\emptyset \in Res^*(E(\varphi))$
- ▶ By Compactness Theorem of propositional logic, there is some finite subset $F = \{\varphi_1, \dots, \varphi_m\} \subseteq E(\varphi)$ such that $\emptyset \in Res^*(F)$
- ▶ So if $\emptyset \in Res^*(\{\varphi_1, \dots, \varphi_m\})$ for some finite m , we conclude φ is unsatisfiable

Checking Satisfiability of φ

- ▶ If $\emptyset \notin \text{Res}^*(\{\varphi_1, \dots, \varphi_m\})$, then we look at $\text{Res}^*(\{\varphi_1, \dots, \varphi_m, \varphi_{m+1}\})$
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- ▶ $\vdots$
- ▶ If φ is satisfiable, then this procedure will continue.

Wrapping Up

- ▶ We have a method to show that a FOL formula φ is unsatisfiable
- ▶ First, write φ in equality free Skolem normal form
- ▶ Check if $\emptyset \in Res^*(E(\varphi))$, this may take some time
- ▶ There is a more systematic resolution for FOL (ground resolution)
- ▶ We also do not cover a direct undecidability proof for the satisfiability of FOL